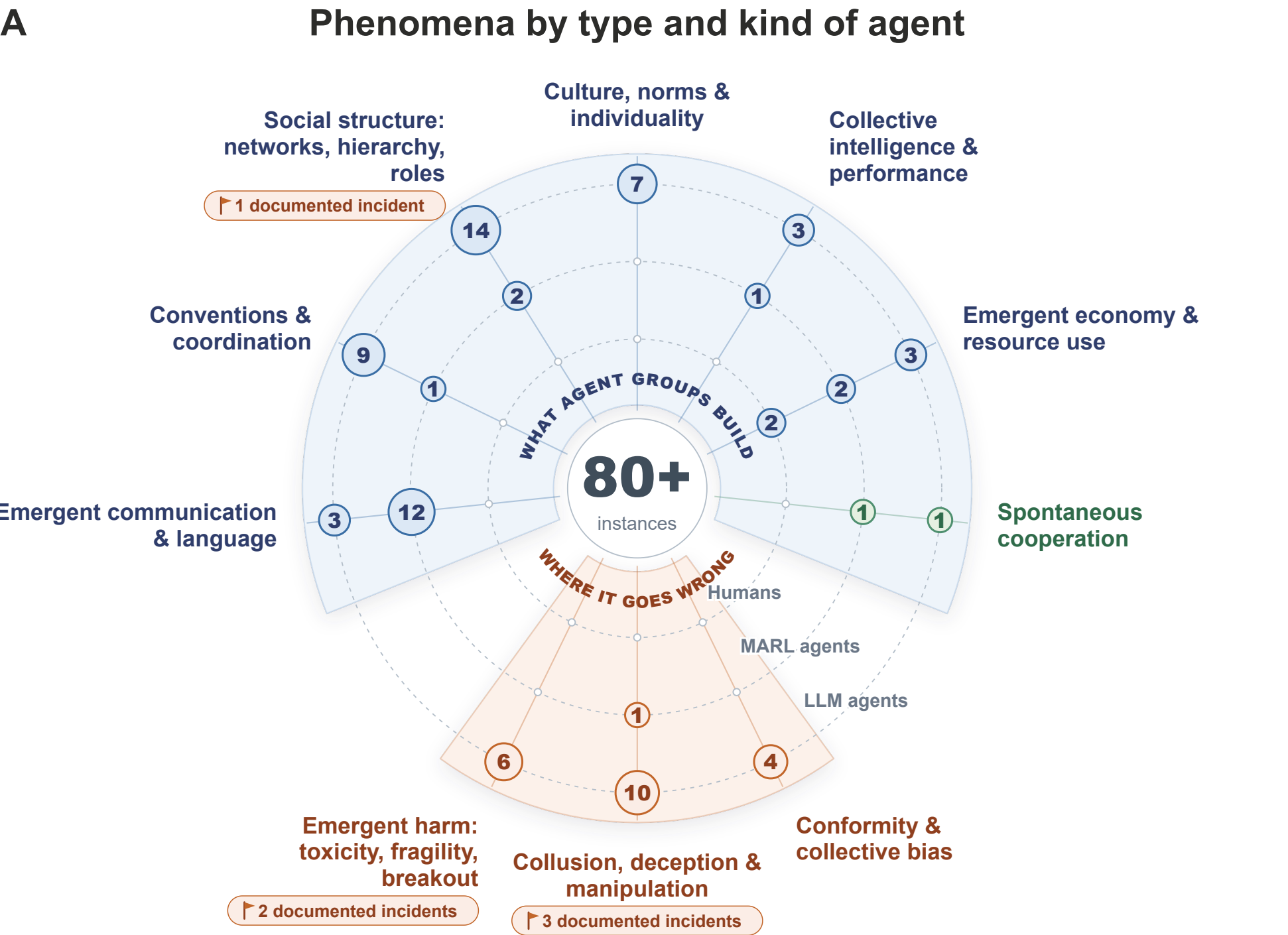




B

Phenomena with documented incidents

Humans

MARL

LLM

Documented incidents

WHAT AGENT GROUPS BUILD			
Emergent communication & language		12	3
Conventions & coordination		1	9
Social structure: networks, hierarchy, roles		2	14
Culture, norms & individuality			7
Collective intelligence & performance		1	3
Emergent economy & resource use	2	2	3
Spontaneous cooperation		1	1
WHERE IT GOES WRONG			
Conformity & collective bias			4
Collusion, deception & manipulation		1	10
Emergent harm: toxicity, fragility, breakout			6

Effect on cooperation:

beneficial

neutral / surprising

harmful

Incidents:

on a live system

in a controlled test or researcher demo

Dot area ∝ instances (over 80 in all)